

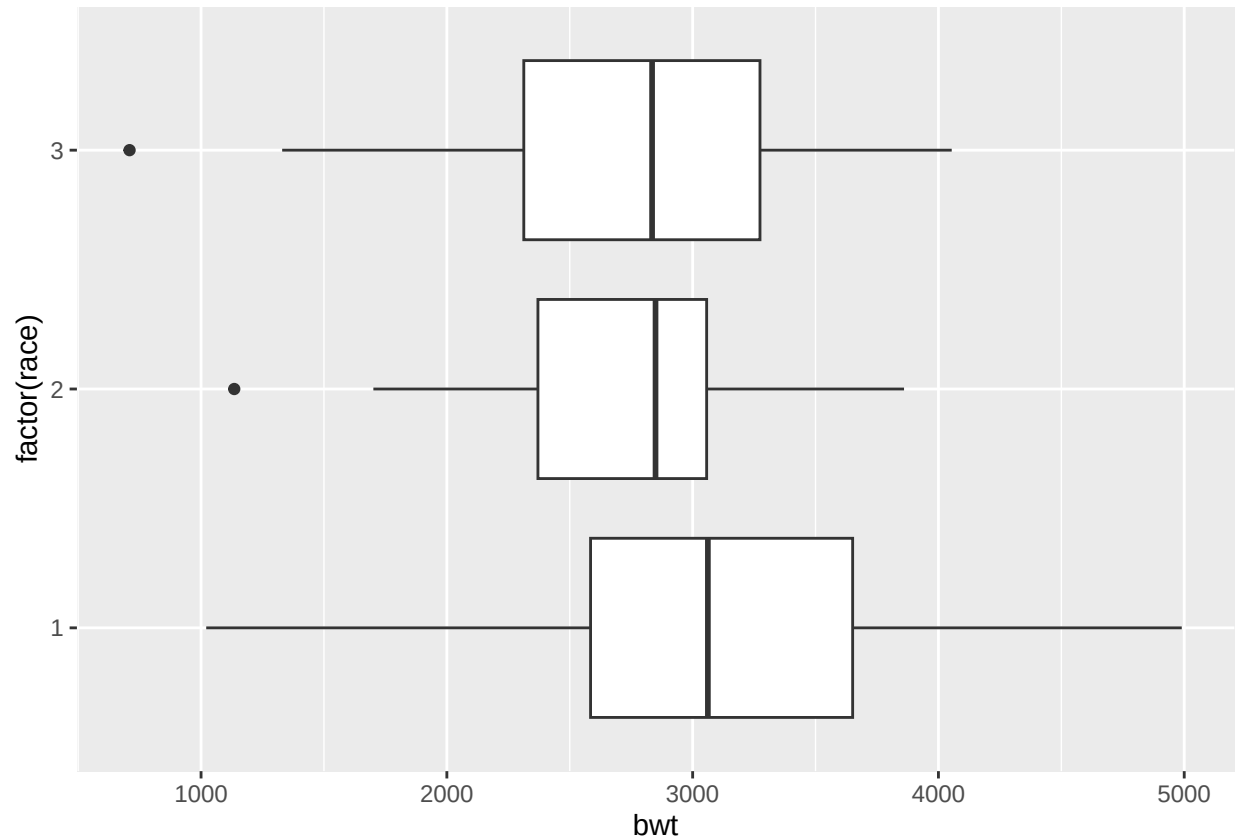
ggplot Functions for Finishing Purposes

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Axes

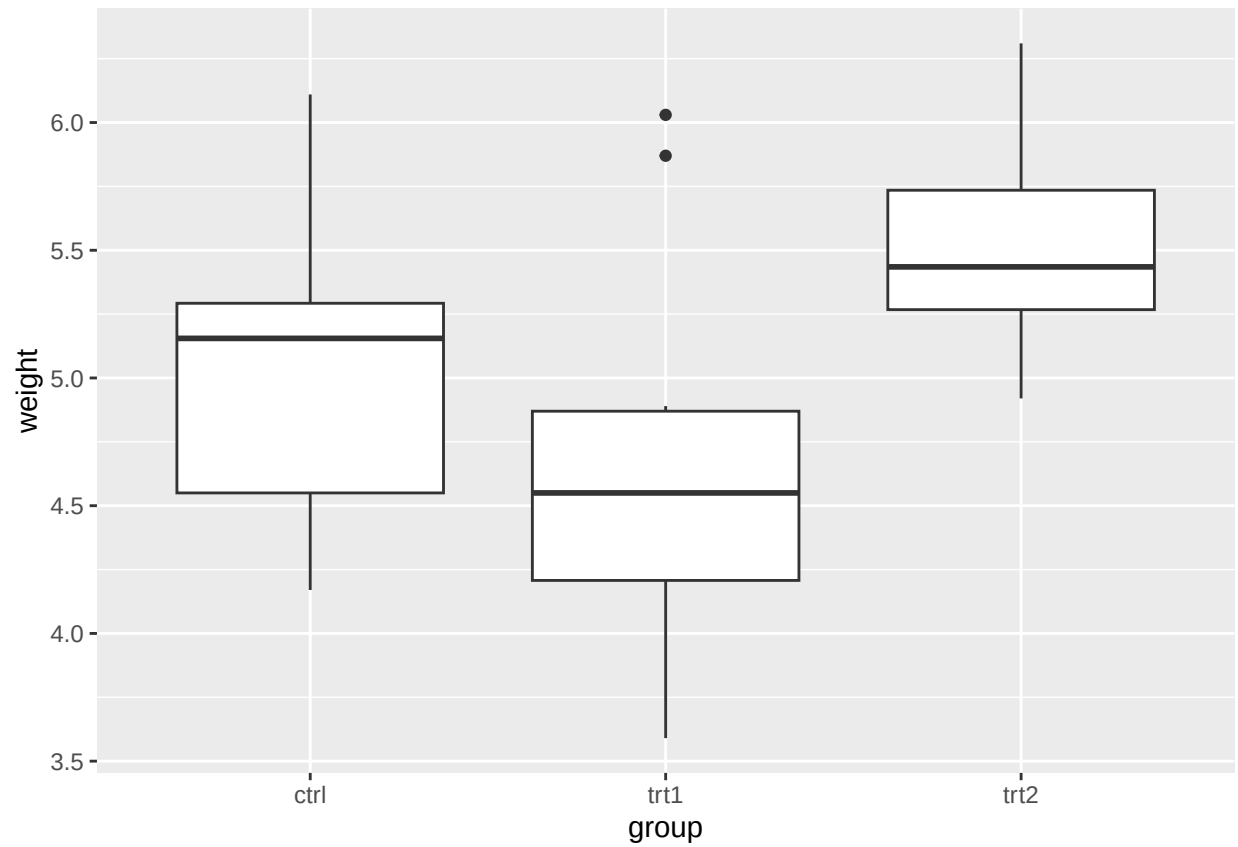
Swap Axes

```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt))  
gg+geom_boxplot()+coord_flip()
```

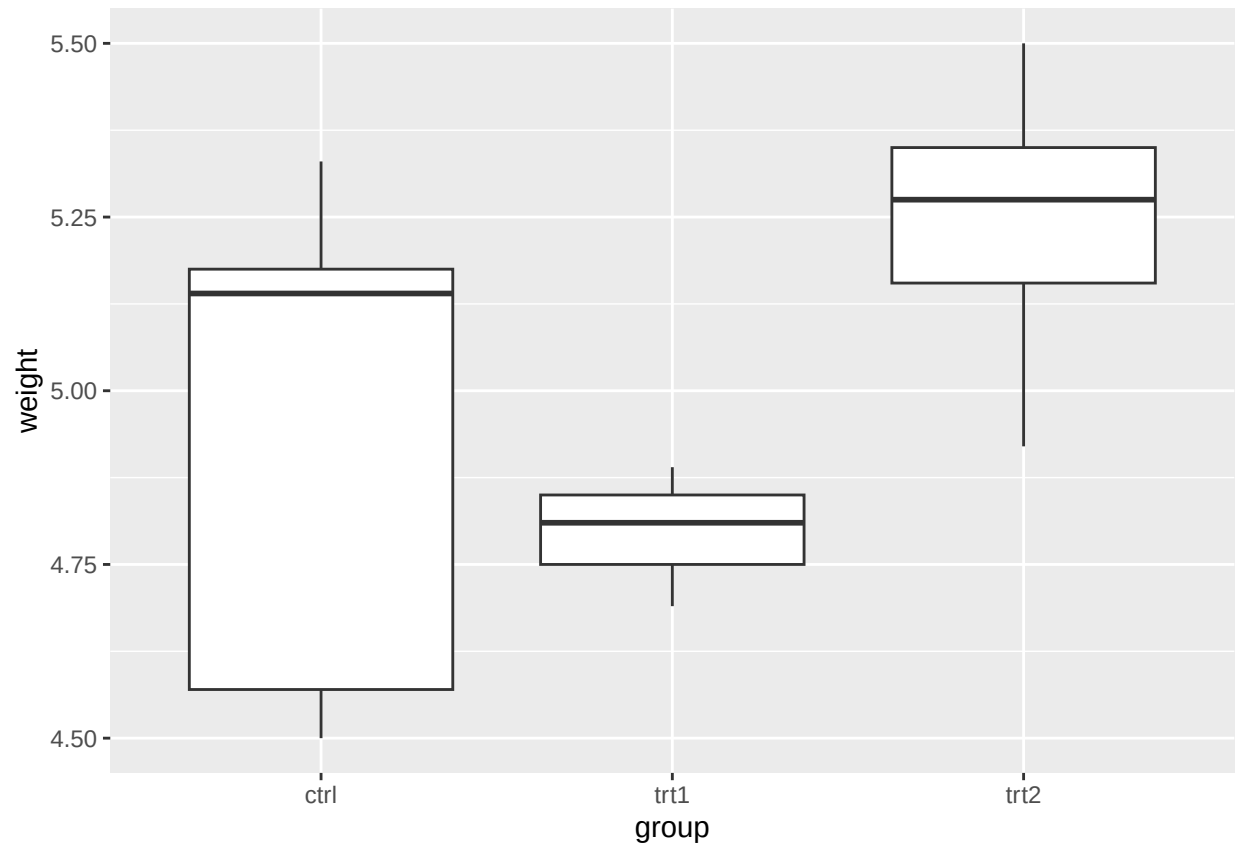


Limits

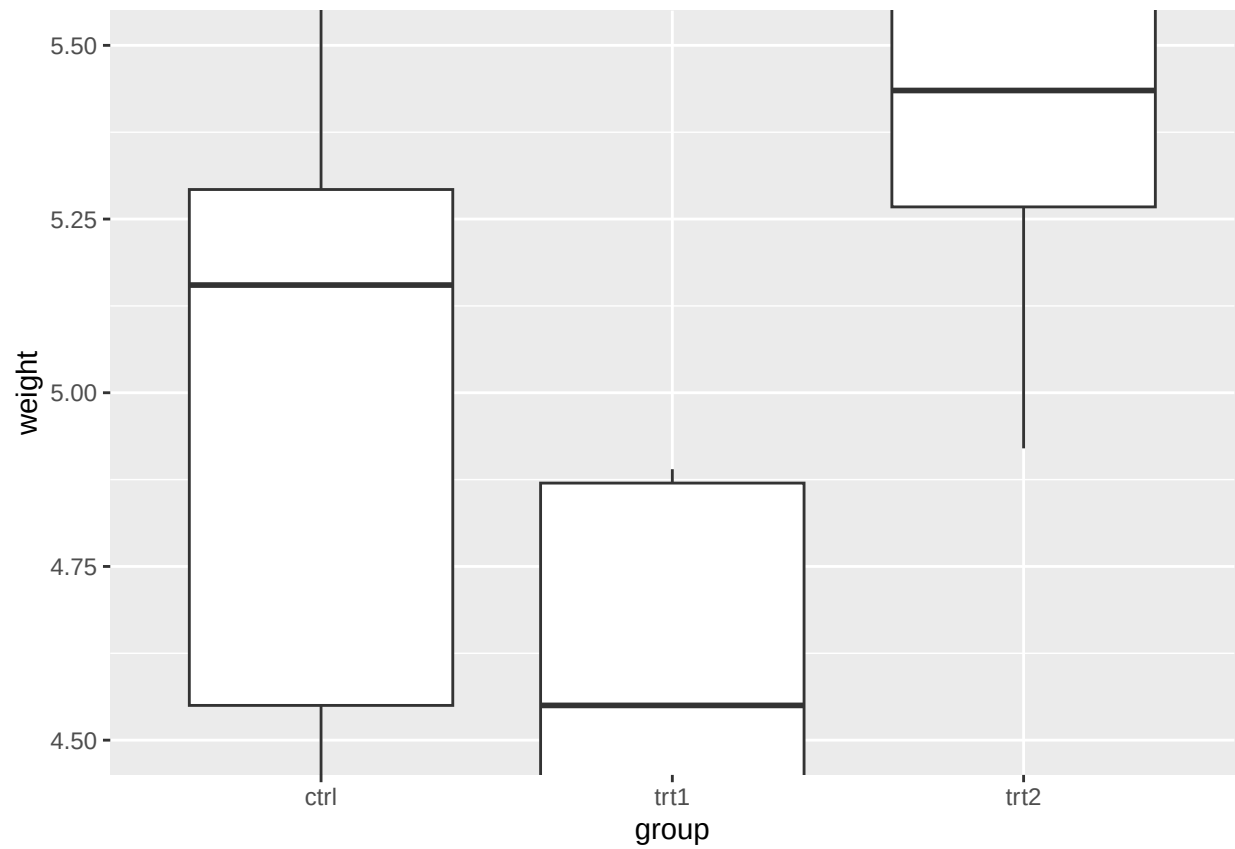
```
ggplot(PlantGrowth,aes(x=group,y=weight))+geom_boxplot()
```



```
ggplot(PlantGrowth,aes(x=group,y=weight))+geom_boxplot()+  
scale_y_continuous(limits=c(4.5,5.5))
```

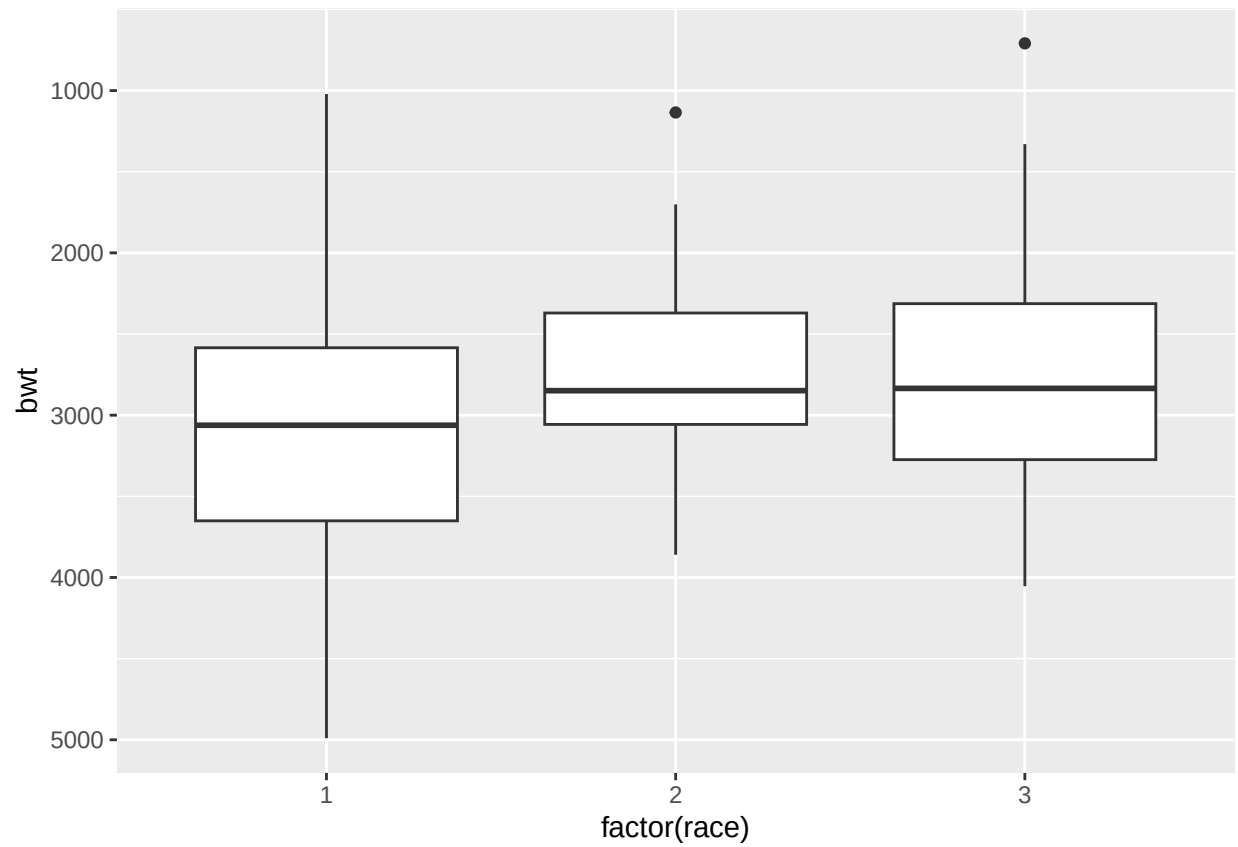


```
ggplot(PlantGrowth,aes(x=group,y=weight))+geom_boxplot()+  
coord_cartesian(ylim=c(4.5,5.5))
```

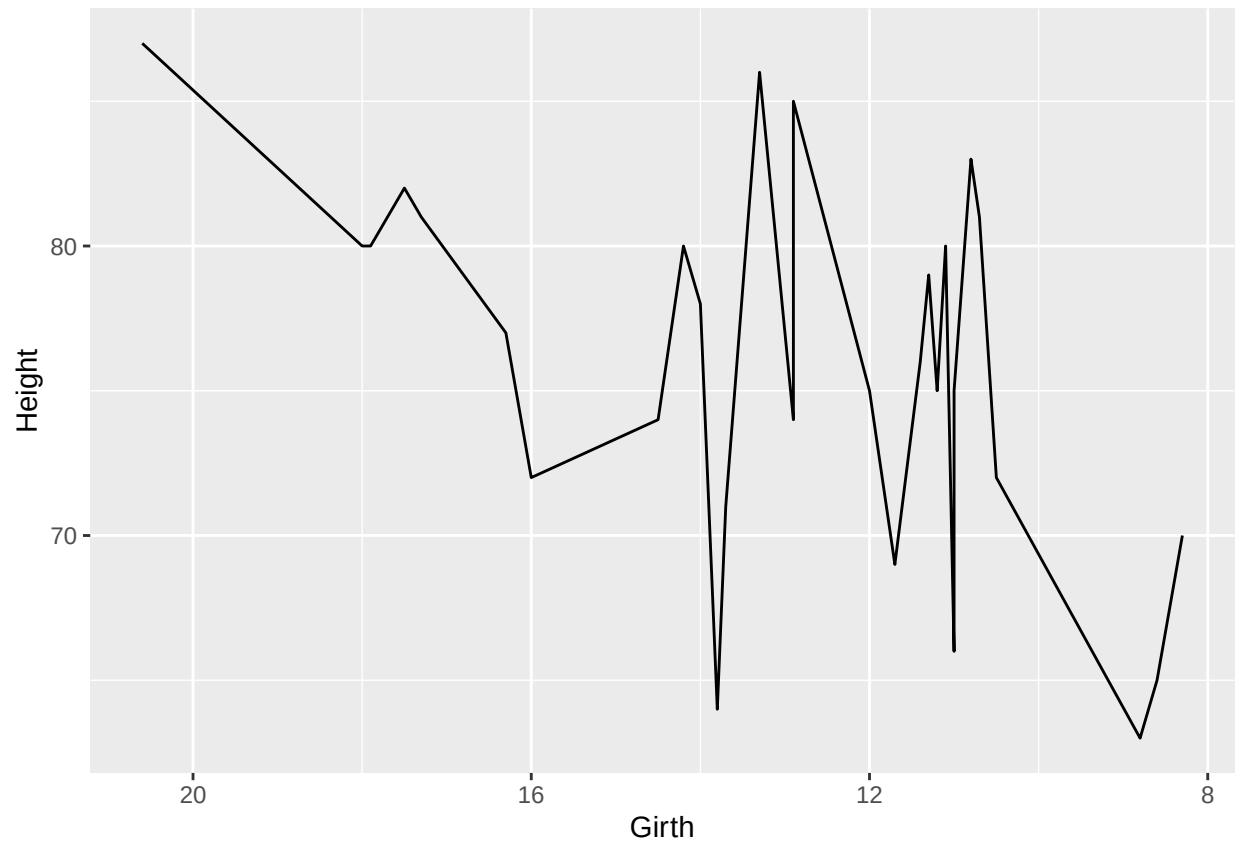


Reverse Axes - Continuous

```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt))  
gg+geom_boxplot()+scale_y_reverse()
```

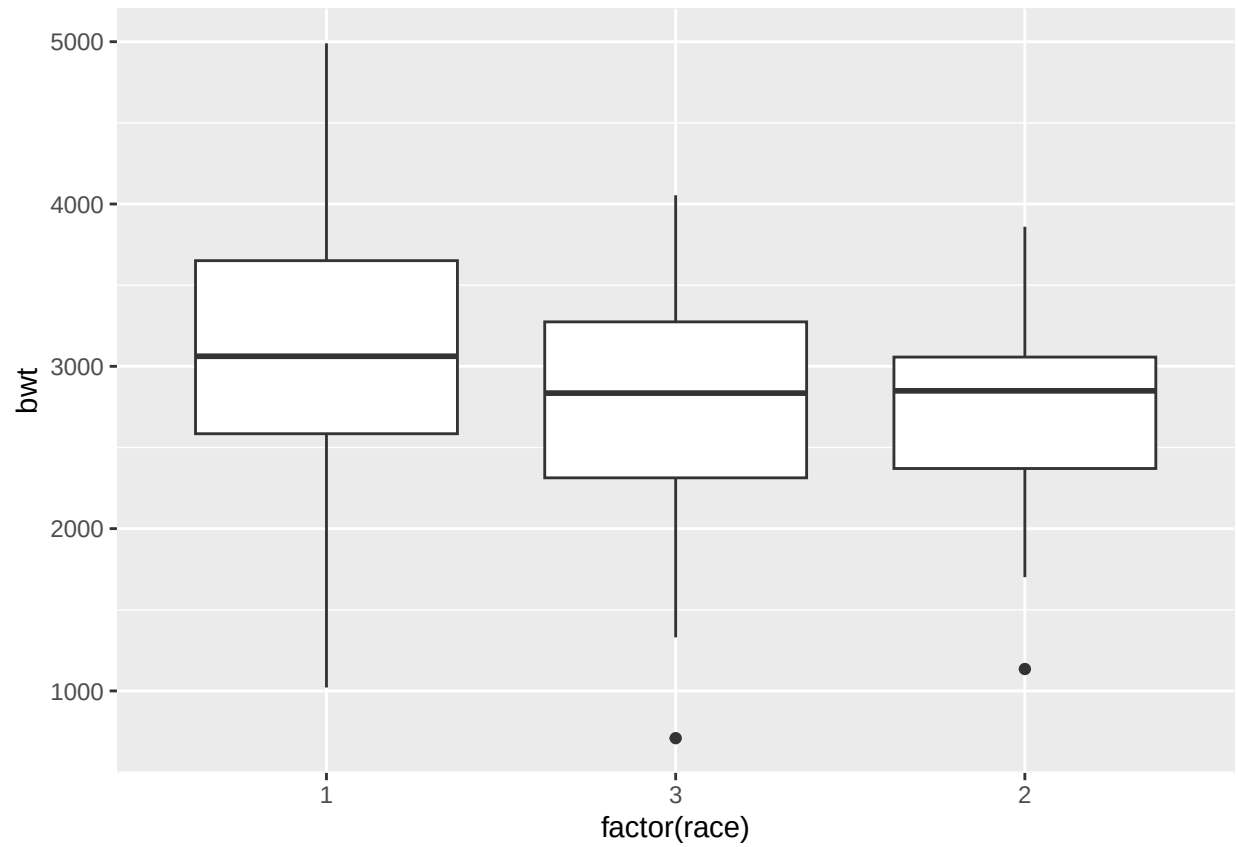


```
ggplot(trees,aes(x=Girth,y=Height))+geom_line()+  
scale_x_reverse()
```

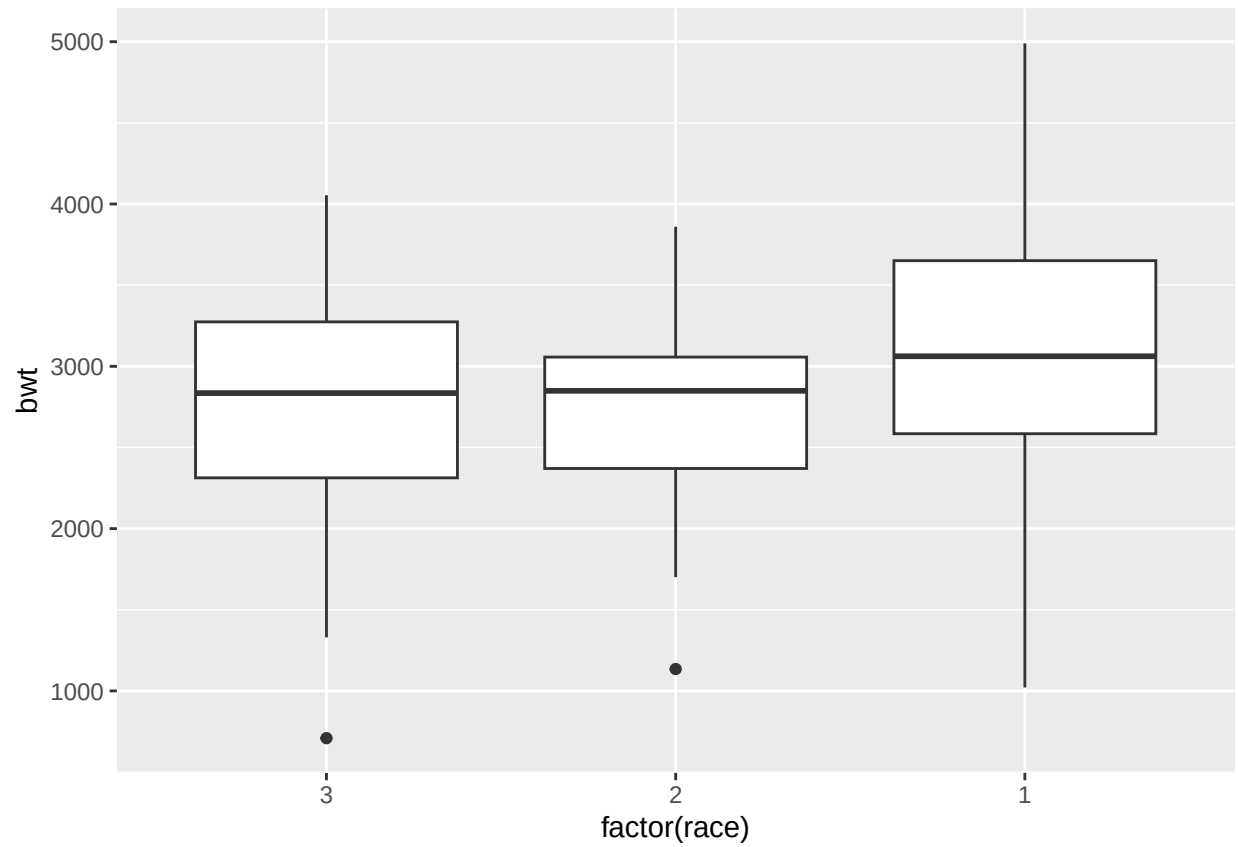


Reorder Items - Discrete

```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt))  
gg+geom_boxplot()+  
scale_x_discrete(limits=as.character(c(1,3,2)))
```



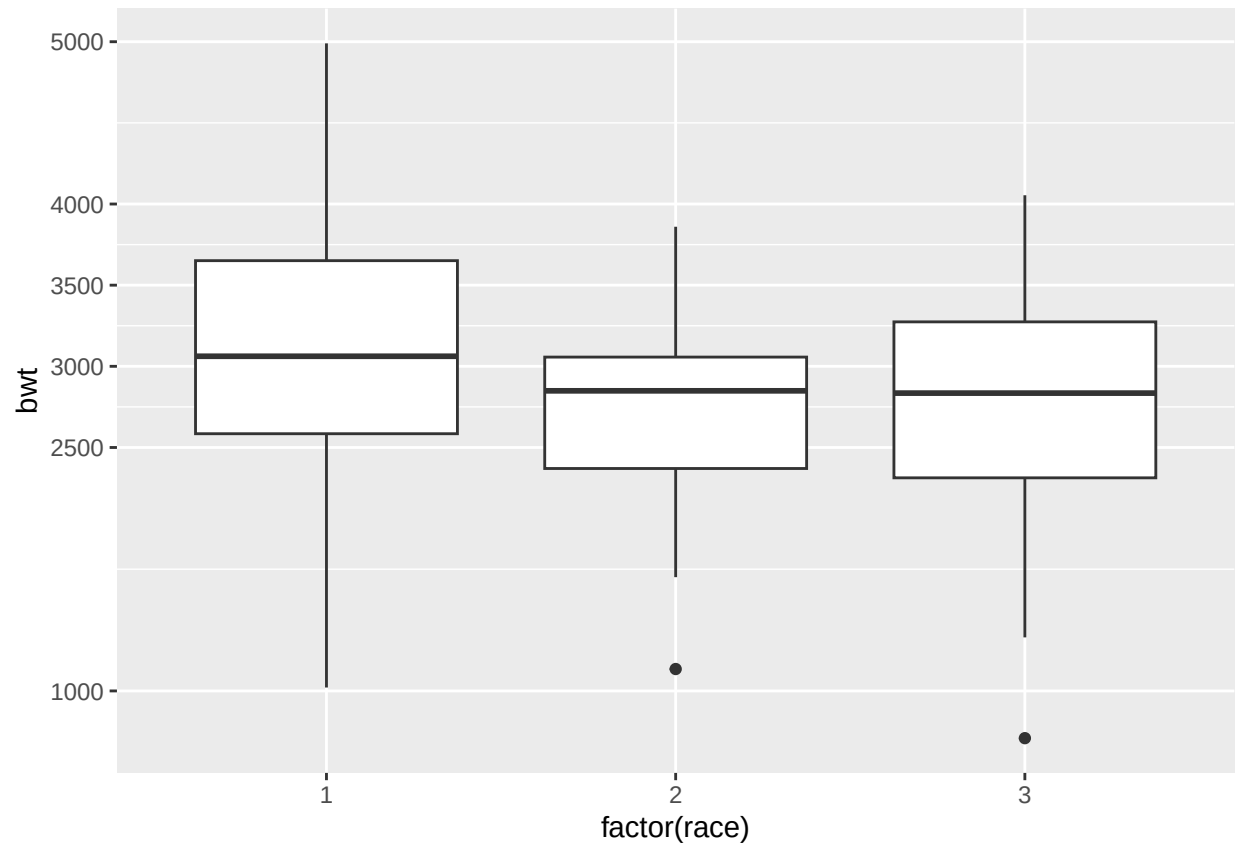
```
gg+geom_boxplot()+  
scale_x_discrete(limits=rev(levels(factor(birthwt$race))))
```



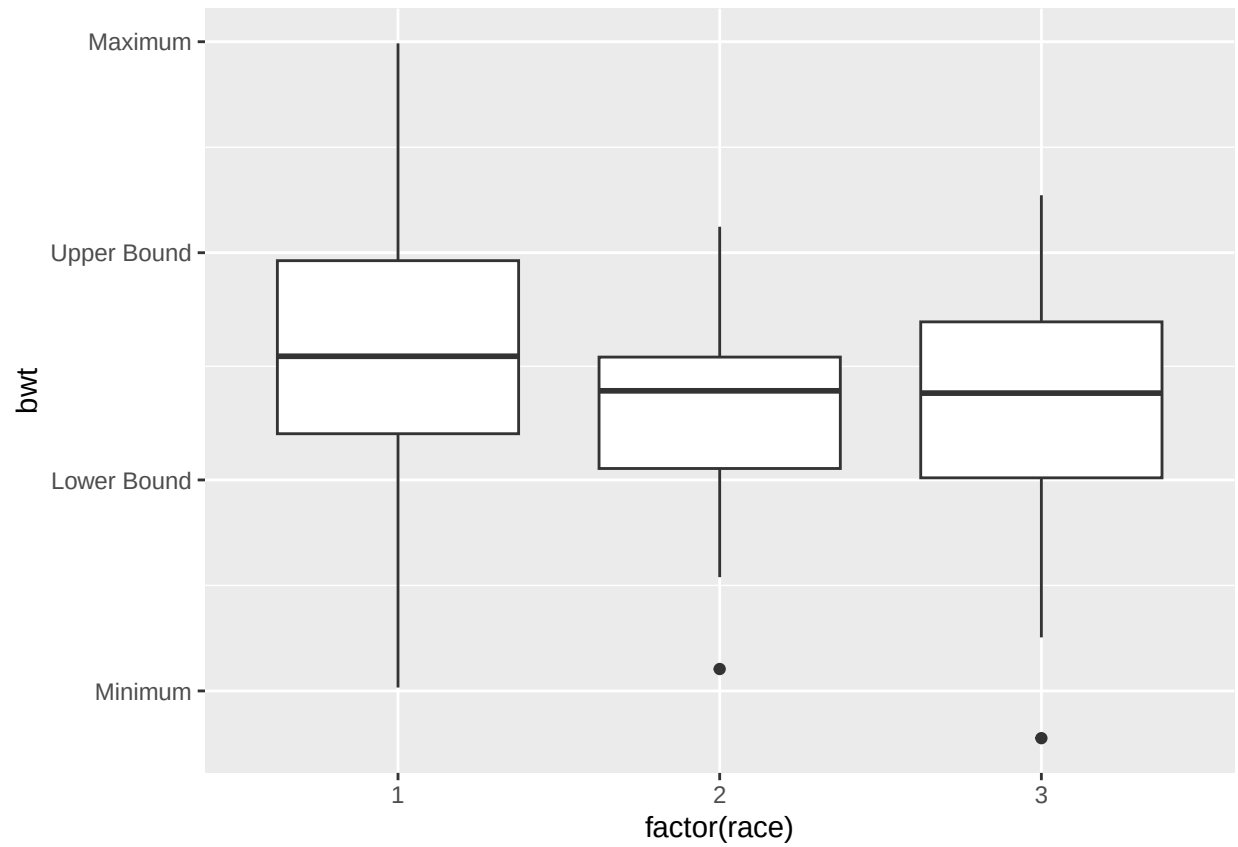
Axes

Tick Marks

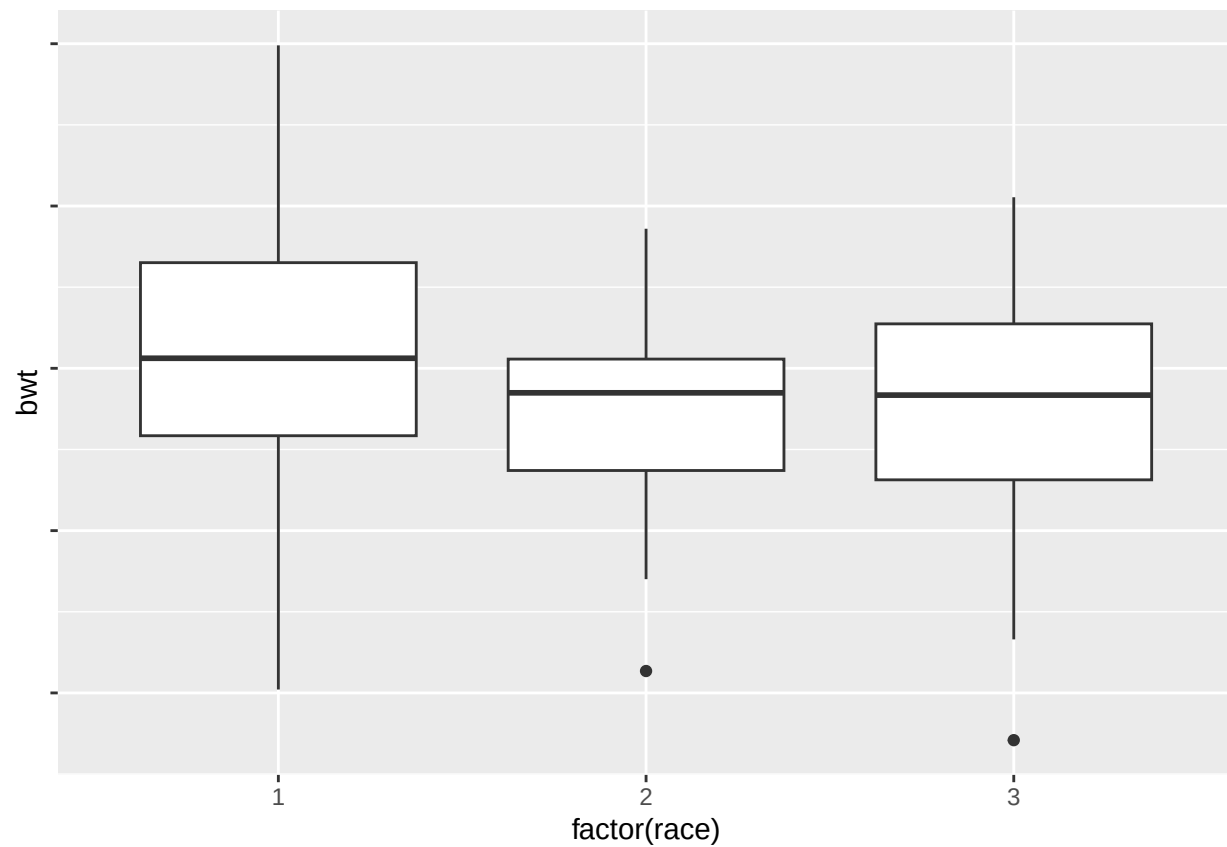
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt))
gg+geom_boxplot()+
scale_y_continuous(breaks=c(1000,2500,3000,3500,4000,5000))
```

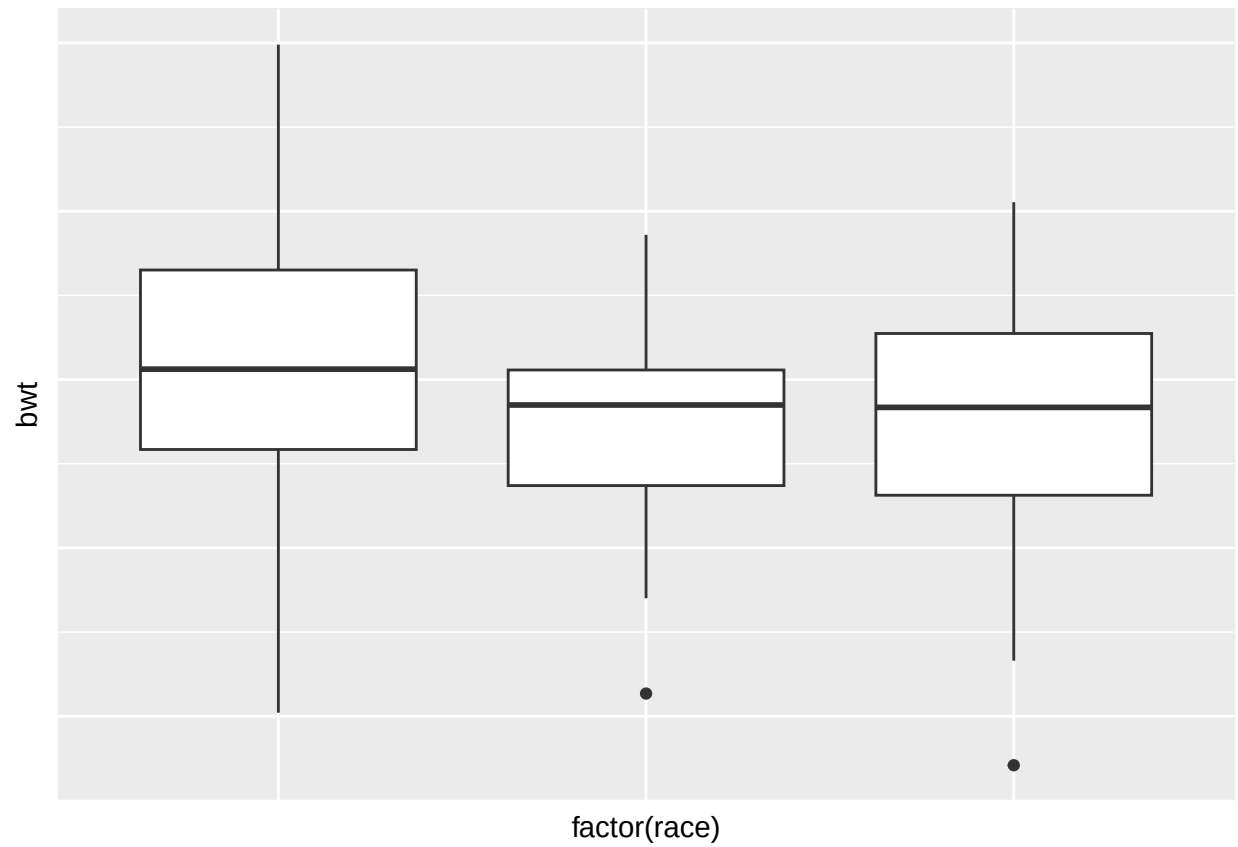
```
gg+geom_boxplot()+  
scale_y_continuous(breaks=c(1000,2300,3700,5000),  
labels=c("Minimum","Lower Bound","Upper Bound","Maximum"))
```



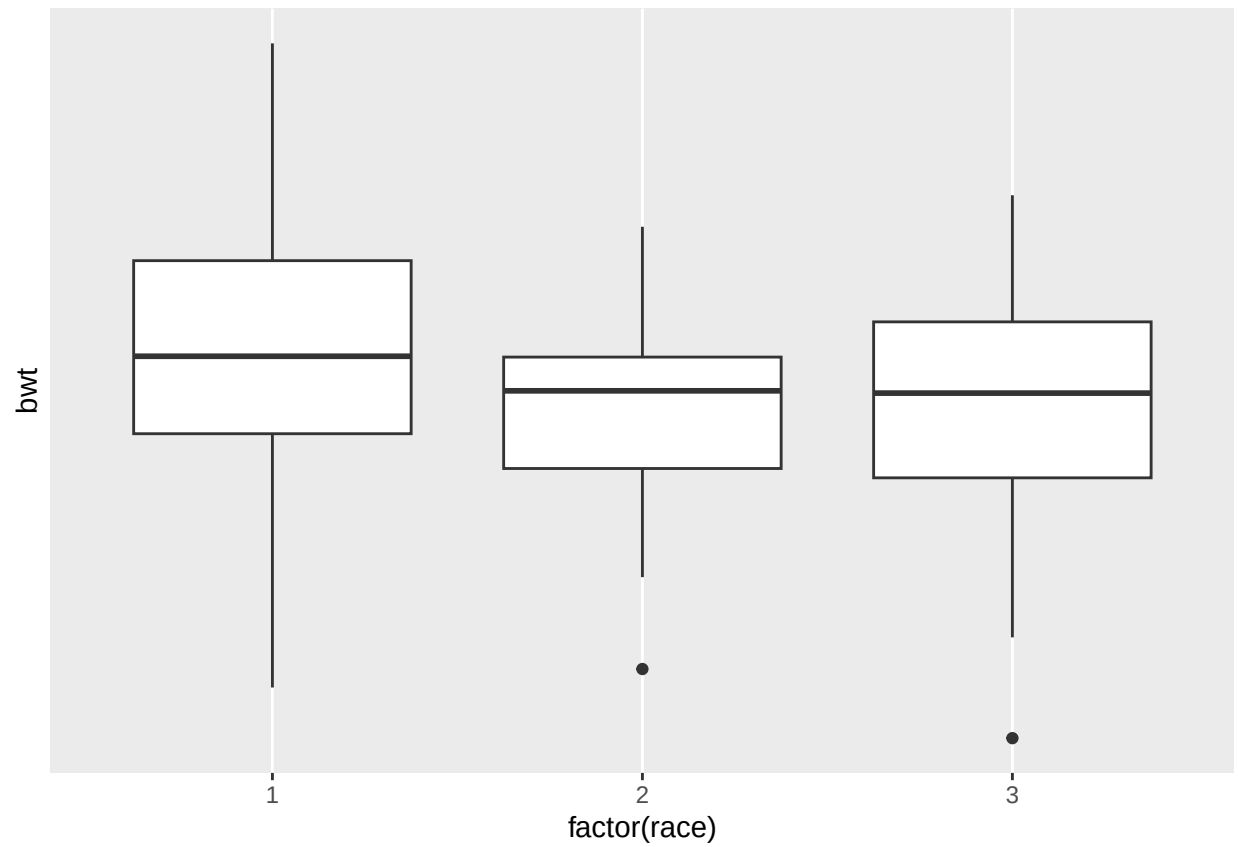
```
gg+geom_boxplot()+theme(axis.text.y=element_blank())
```



```
gg+geom_boxplot()+theme(axis.ticks=element_blank(),  
axis.text=element_blank())
```



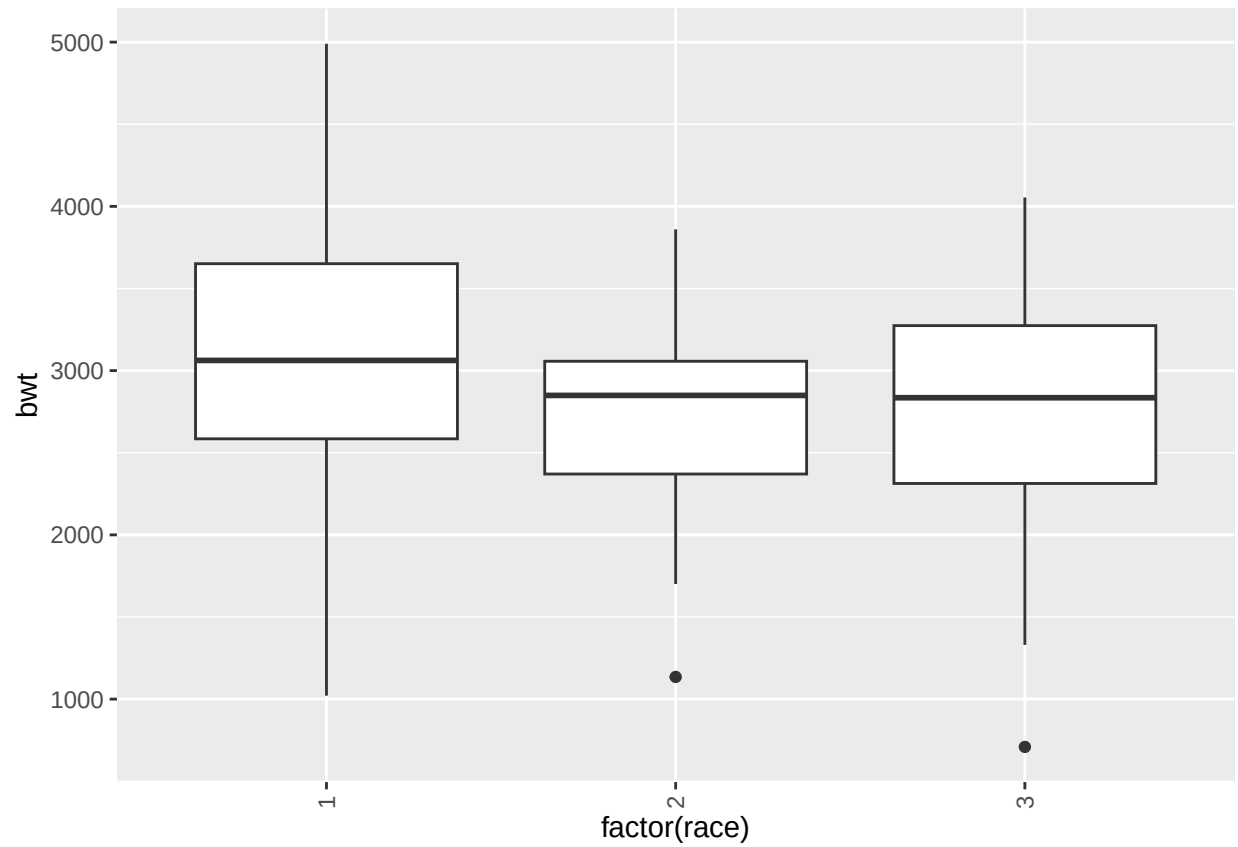
```
gg+geom_boxplot()+scale_y_continuous(breaks=NULL)
```



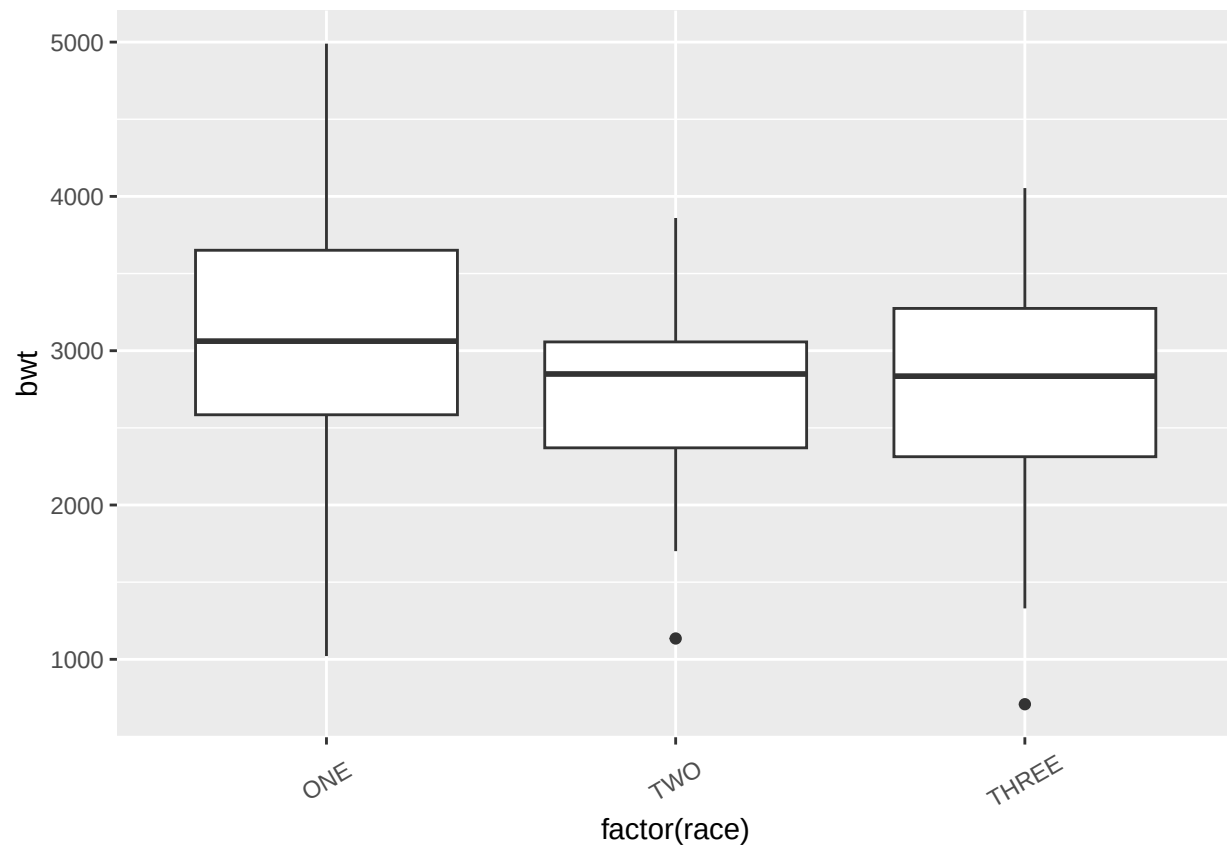
Axes

Text on Axes

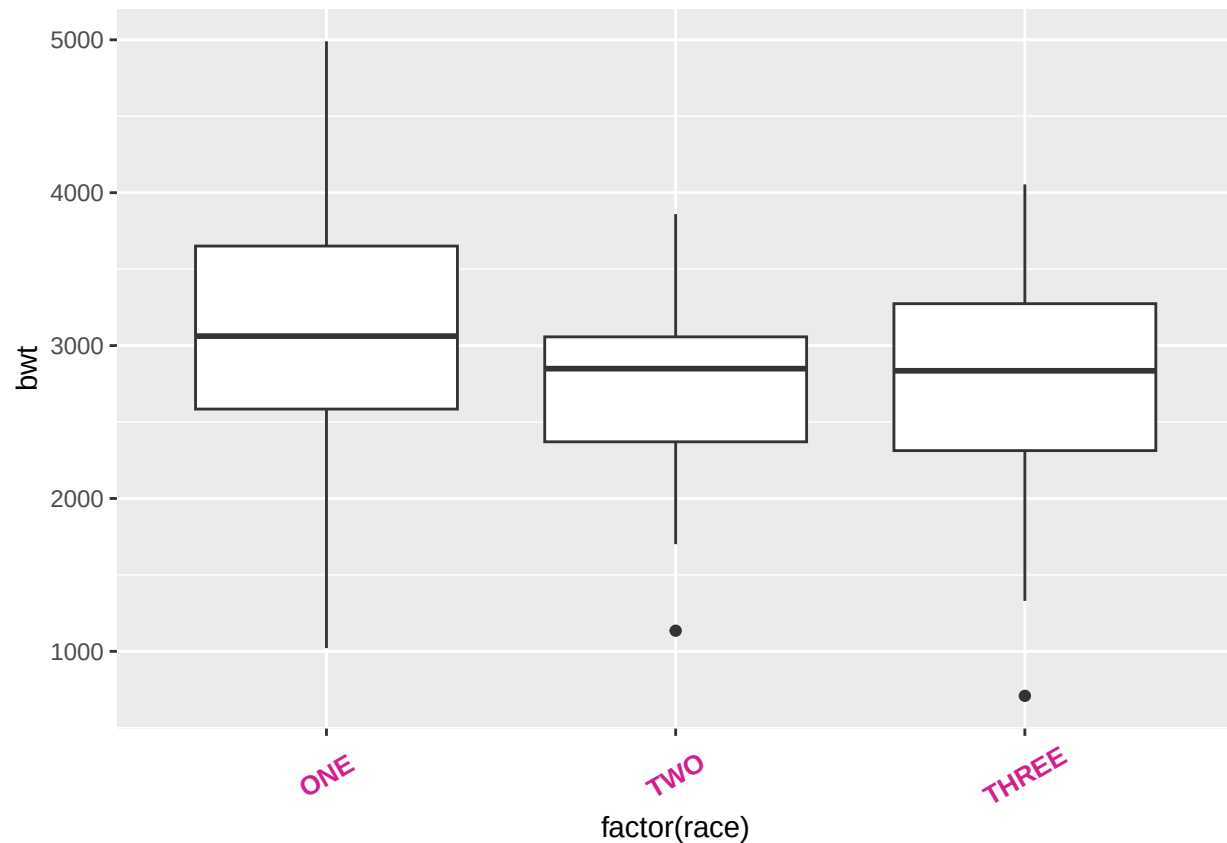
```
gg<-ggplot(birthwt,aes(x=factor(race),y=bwt))
gg+geom_boxplot()+
theme(axis.text.x=element_text(angle=90,hjust=0.5,vjust=0.5))
```



```
gg+geom_boxplot()+scale_x_discrete(labels=c("ONE","TWO","THREE"))+  
theme(axis.text.x=element_text(angle=30,hjust=0.5,vjust=0.5))
```



```
gg+geom_boxplot()+scale_x_discrete(labels=c("ONE","TWO","THREE"))+  
theme(axis.text.x=element_text  
(angle=30,hjust=0.5,vjust=0.5,colour="violetred",face="bold",size=10))
```



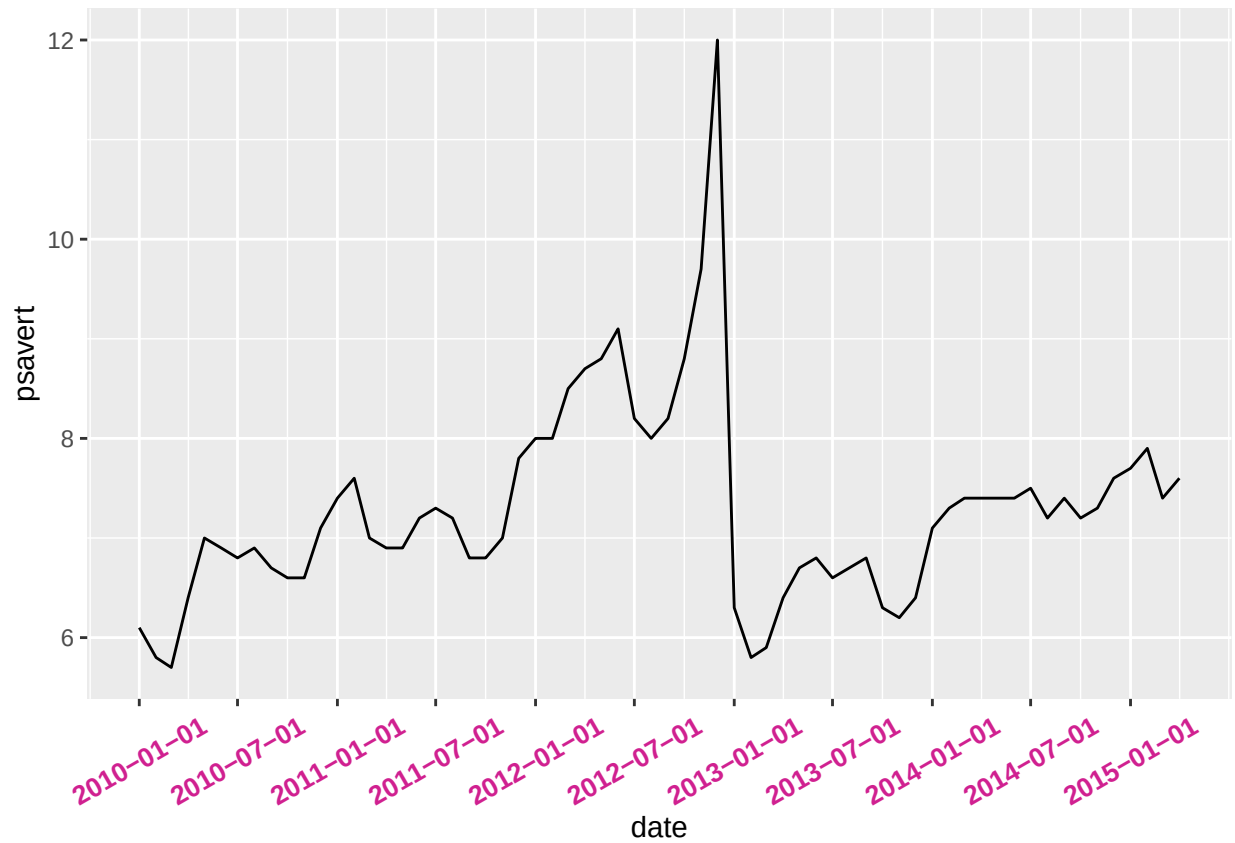
Axes

Date as Axis

```
esub<-subset(economics,date>=as.Date("2010-01-01"))
db<-seq(as.Date("2010-01-01"),as.Date("2015-04-01"),by="6 month")
db

## [1] "2010-01-01" "2010-07-01" "2011-01-01" "2011-07-01" "2012-01-01"
## [6] "2012-07-01" "2013-01-01" "2013-07-01" "2014-01-01" "2014-07-01"
## [11] "2015-01-01"

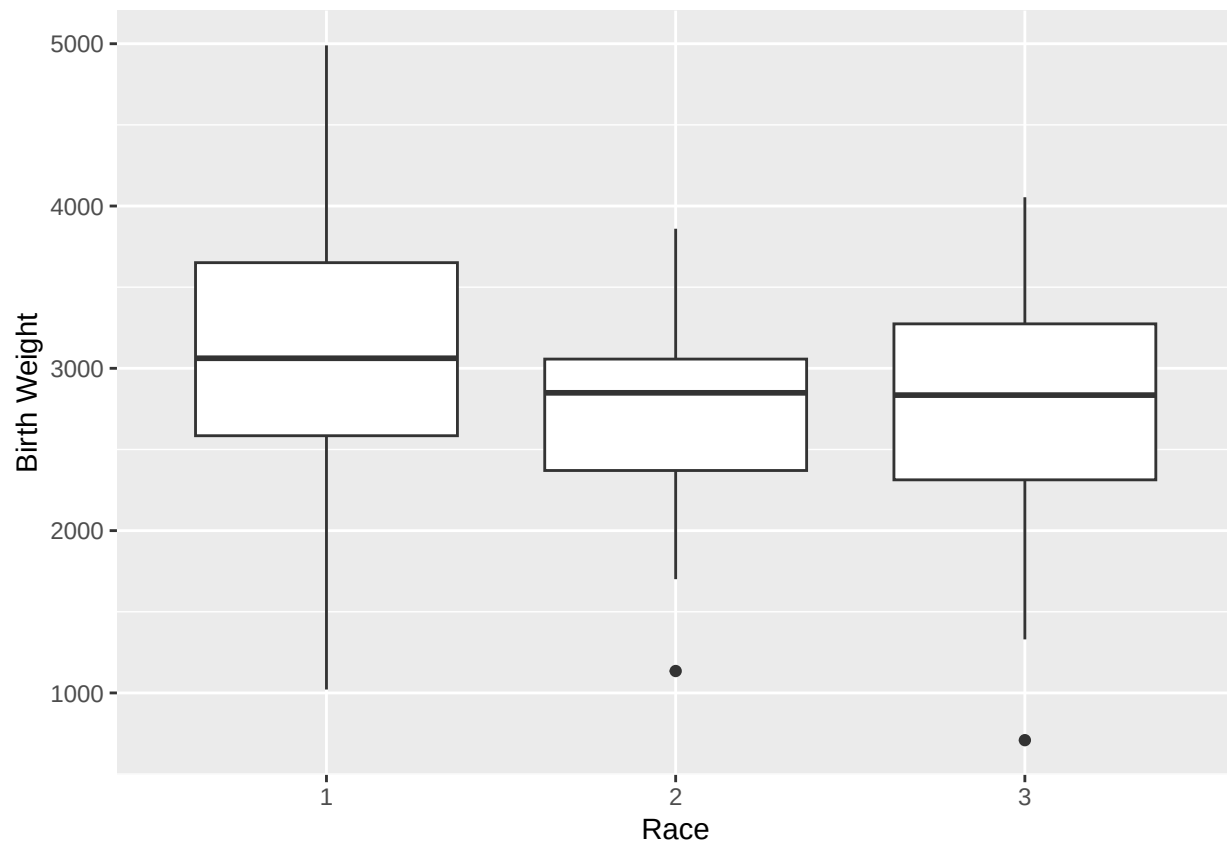
ggplot(esub,aes(x=date,y=psavert))+geom_line()+
scale_x_date(breaks=db)+
theme(axis.text.x=element_text
(angle=30,hjust=0.5,vjust=0.5,colour="violetred",face="bold",size=10))
```

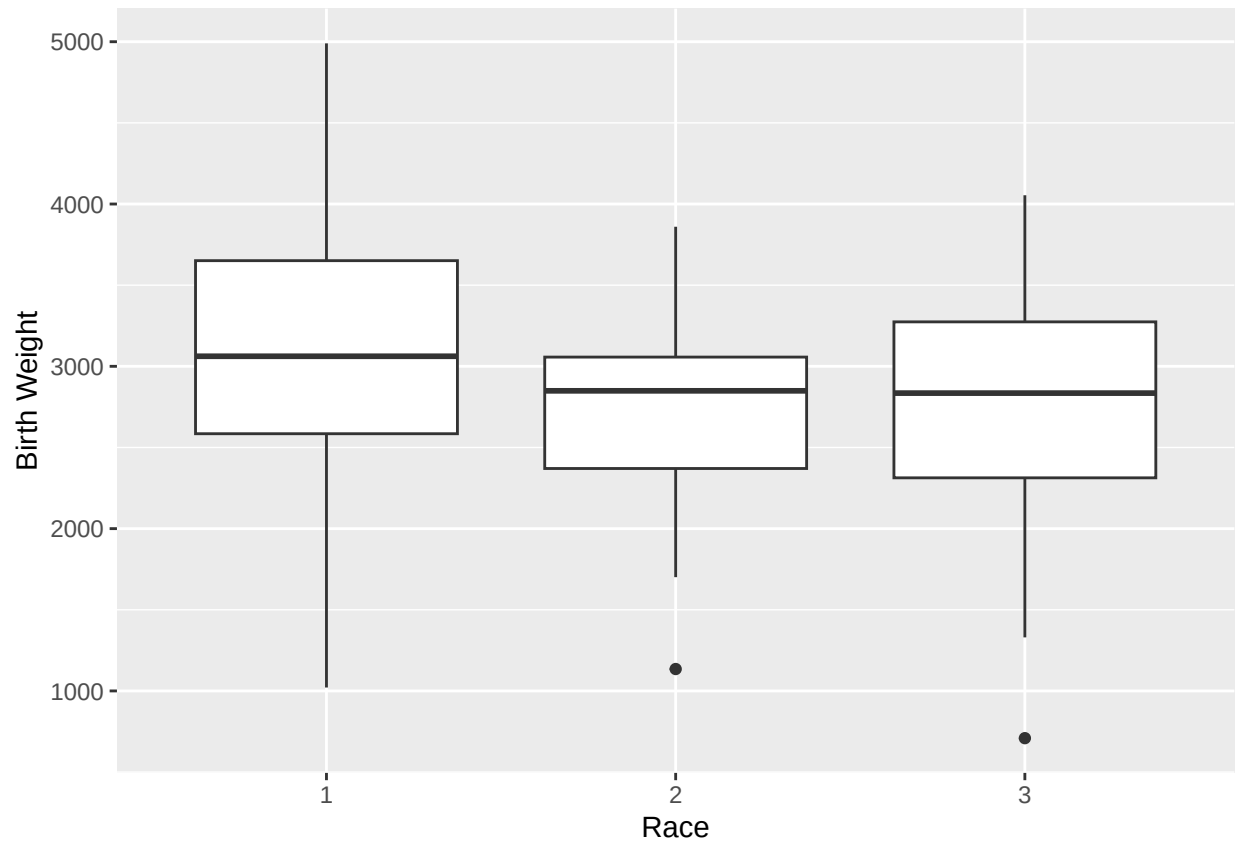
Labels

Change Labels

```
gg<-ggplot(birthwt,aes(x=as.factor(race),y=bwt))  
gg+geom_boxplot()+xlab("Race")+ylab("Birth Weight")
```

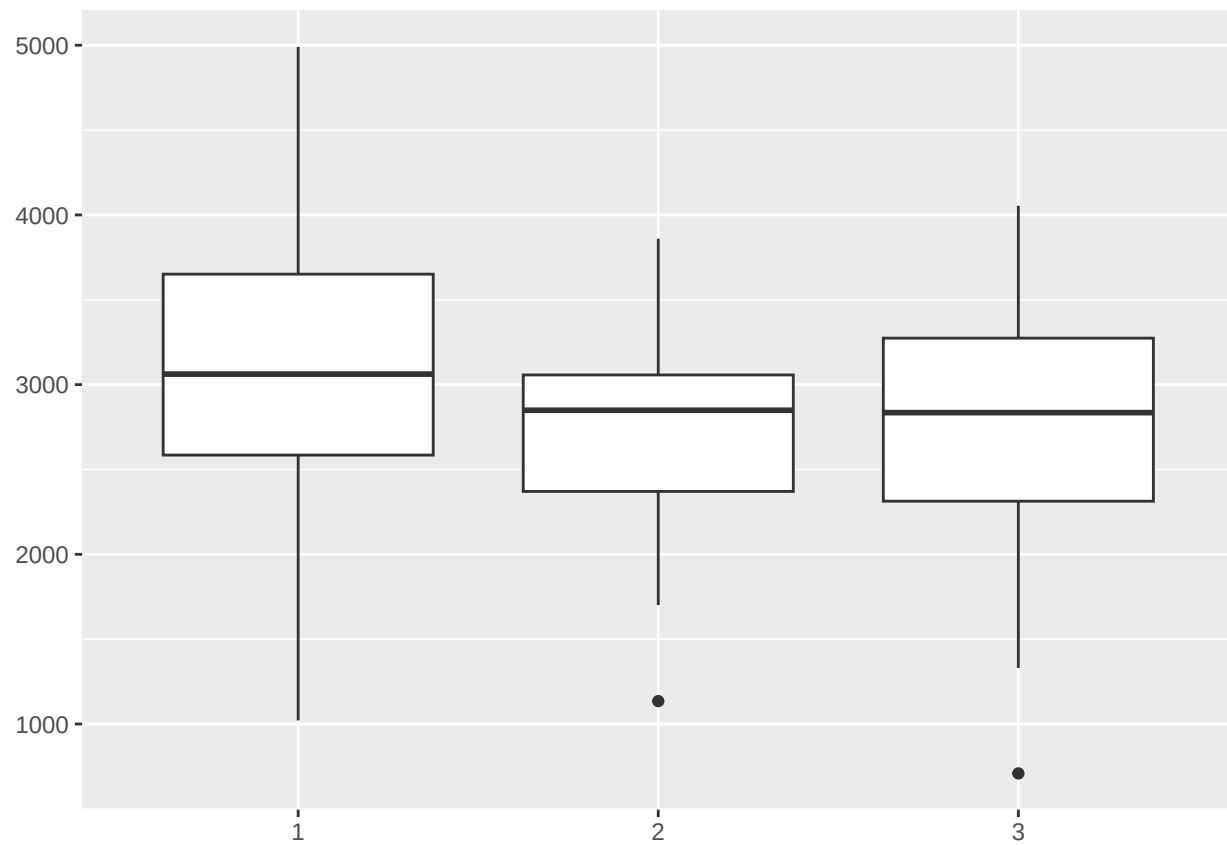


```
gg+geom_boxplot()+scale_x_discrete(name="Race")+  
scale_y_continuous(name="Birth Weight")
```

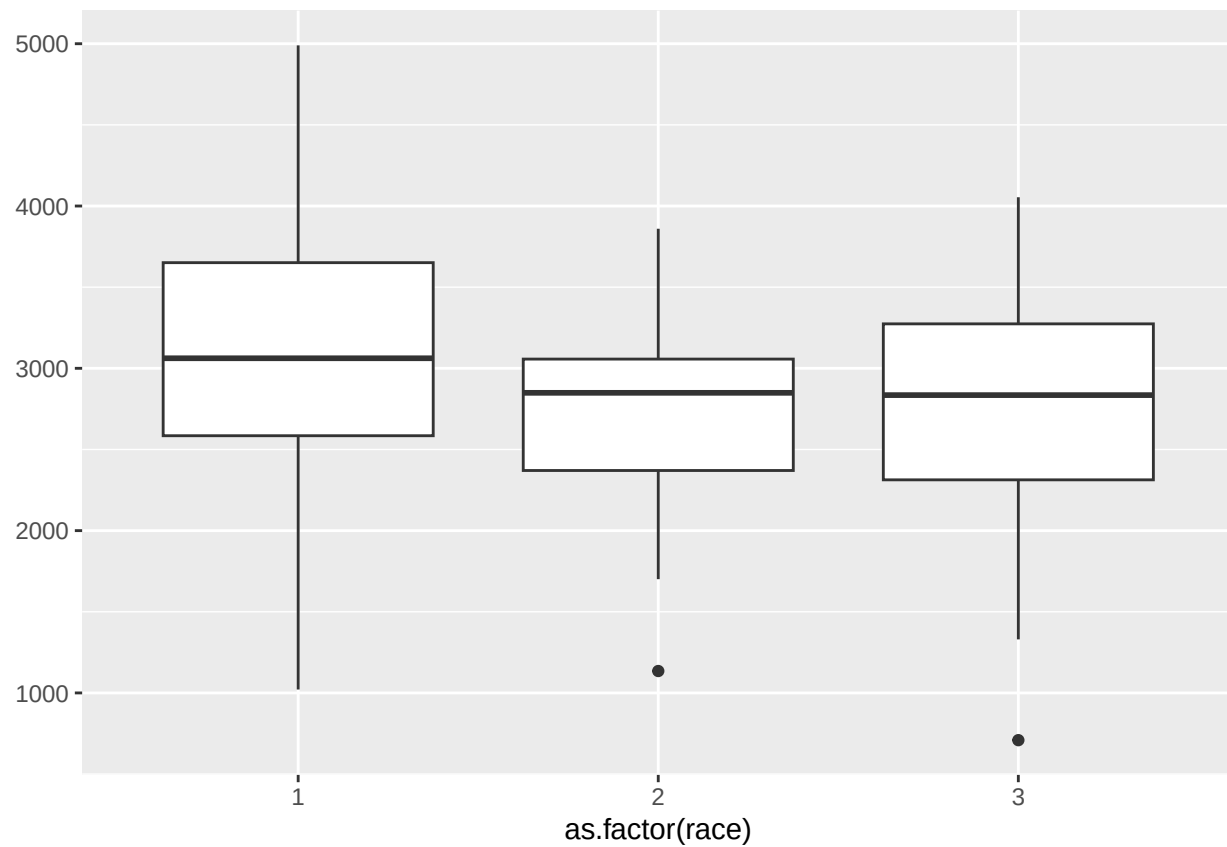


Remove Labels

```
gg<-ggplot(birthwt,aes(x=as.factor(race),y=bwt))  
gg+geom_boxplot()+theme(axis.title=element_blank())
```

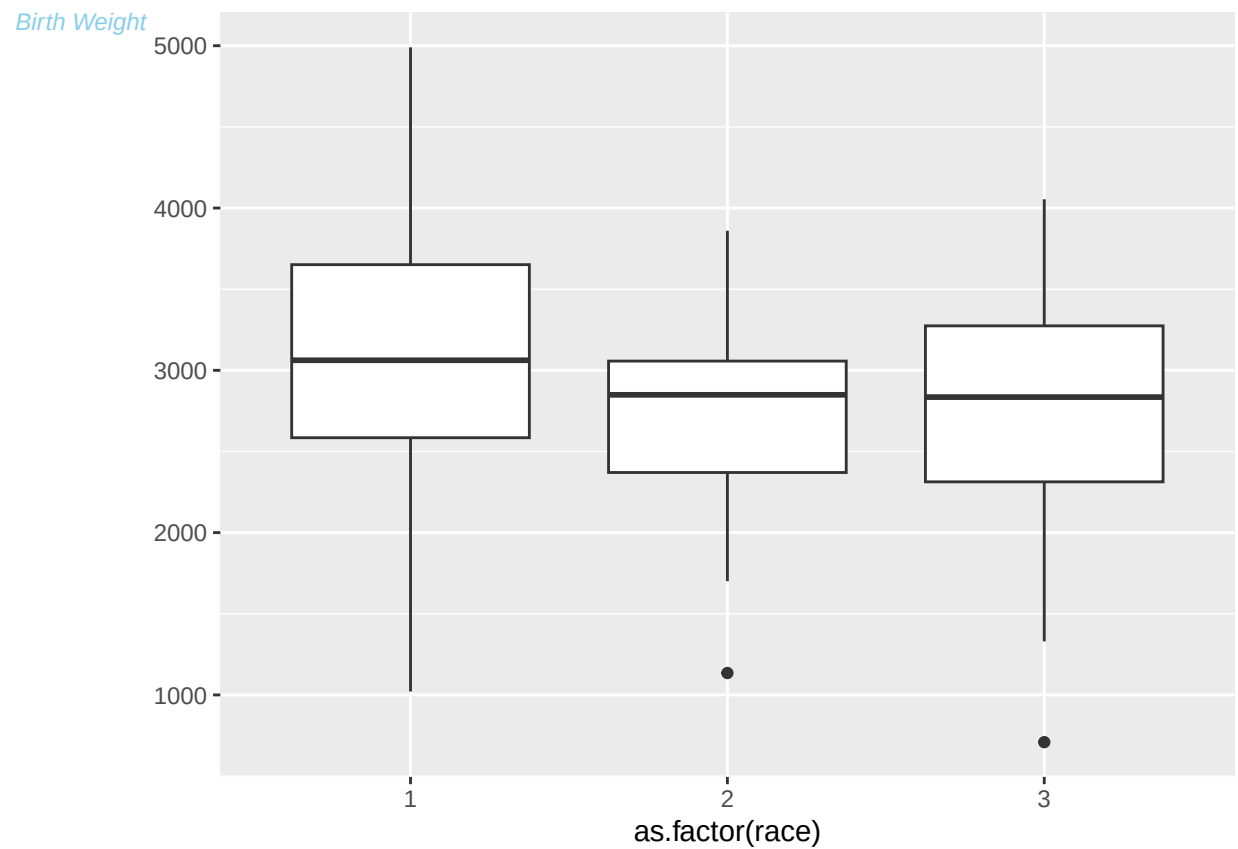


```
gg+geom_boxplot()+theme(axis.title.y=element_blank())
```



Change Appearance

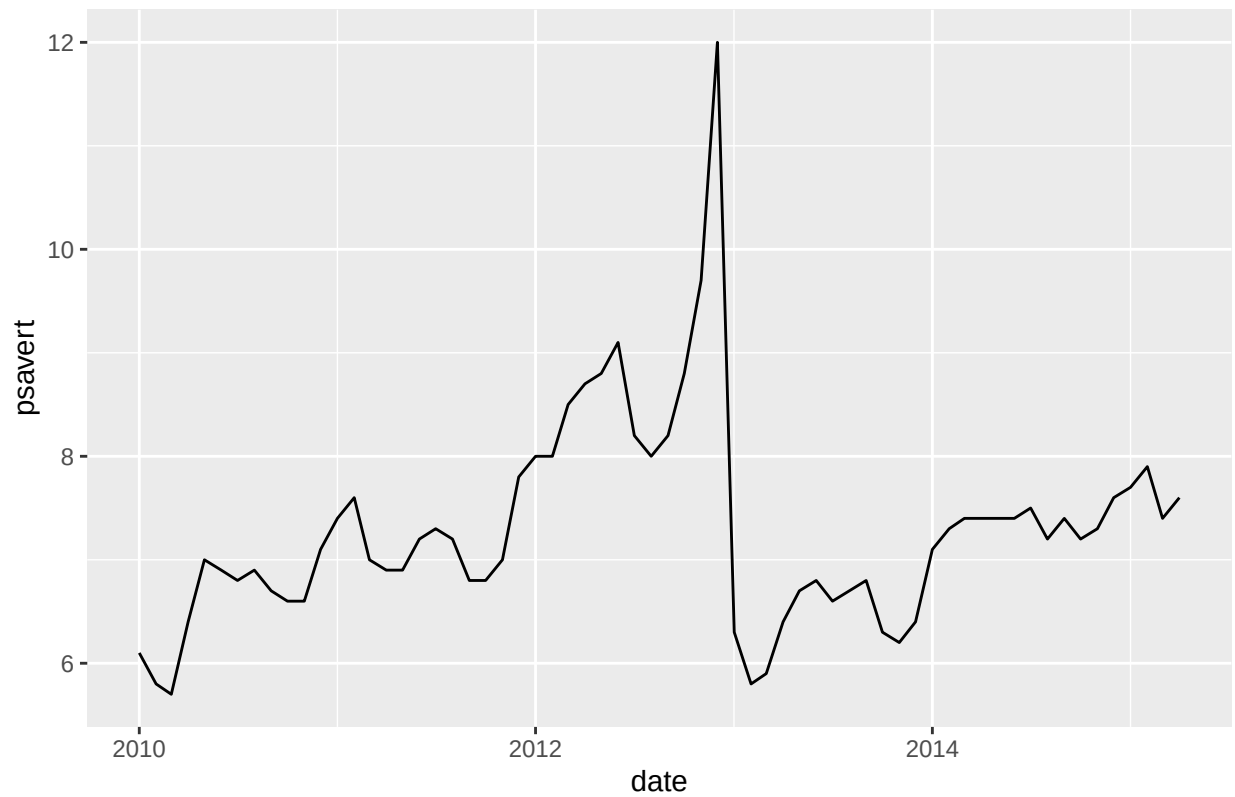
```
gg<-ggplot(birthwt,aes(x=as.factor(race),y=bwt))
gg+geom_boxplot()+ylab("Birth Weight")+
theme(axis.title.y=element_text(
face="italic",angle=0,size=9,colour="skyblue"))
```



Title

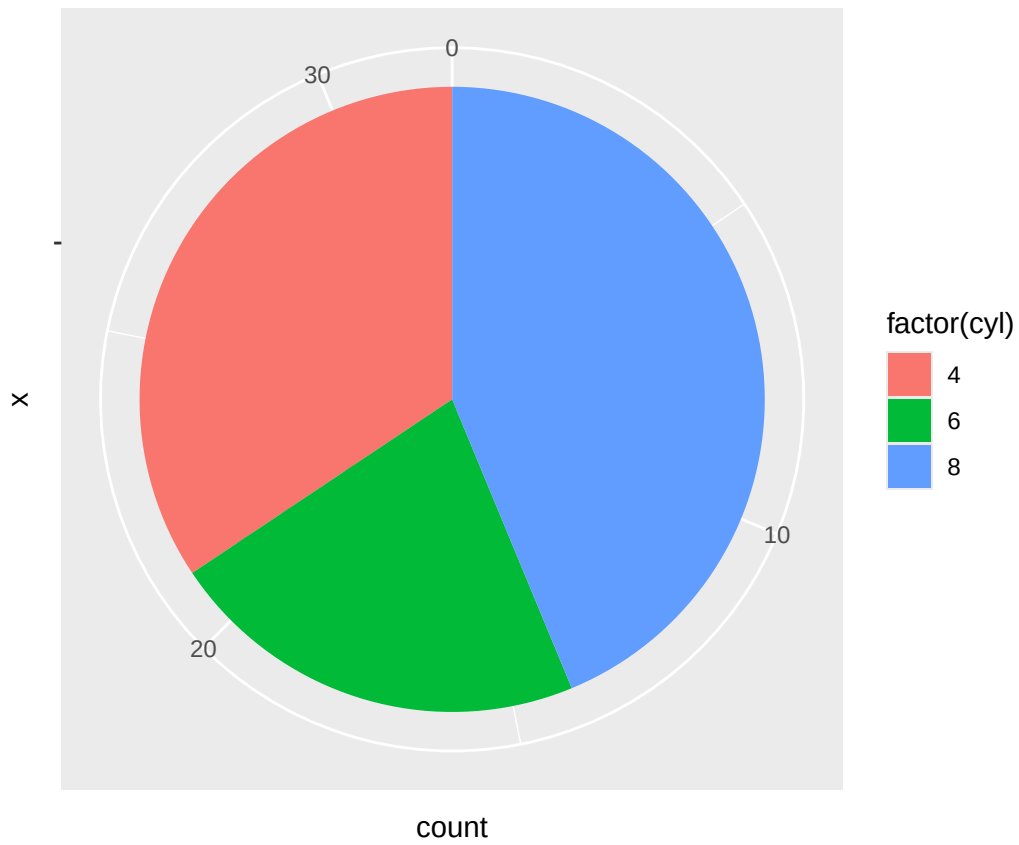
```
ggplot(esub,aes(x=date,y=psavert))+geom_line()+  
ggtitle("Personal Savings Rate 2010 - 2015")+  
theme(plot.title=element_text(face="bold.italic",colour="darkred"))
```

Personal Savings Rate 2010 – 2015



Circular Graph

```
pie <- ggplot(mtcars, aes(x = " ", fill = factor(cyl)))  
pie + geom_bar(width=1)+coord_polar(theta = "y")
```



```
pie + geom_bar(width=1)+coord_polar()
```